Arista Access Node SOP: QoS Congestion Event Resolution on BAS-ACC-78 at Basavanagudi POP

Node: BAS-ACC-78 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_9486

Description

This Standard Operating Procedure (SOP) outlines the steps required to diagnose and resolve Quality of Service (QoS) congestion events on Arista Access nodes, specifically BAS-ACC-78 located at Basavanagudi POP. The failure domain of interest is microbursts, interface tail drops, and QoS queue starvation. The underlying protocol mechanics that contribute to this failure domain are the queuing algorithms used in the egress queues of the Arista Access node. The Priority Flow Control (PFC) protocol and the IEEE 802.1p header are used to classify and prioritize traffic flows. However, in scenarios where there is a sudden surge in traffic, the egress queues can become congested, leading to microbursts and interface tail drops. The QoS queue starvation is caused by the lack of available buffer space in the egress queues, resulting in packets being dropped. This SOP will guide the network administrator through the diagnostic and resolution steps required to identify and mitigate the root cause of the QoS congestion event.

Pre-requisite Checks

- Check: Verify that the BAS-ACC-78 node is accessible via SSH or console
- Check: Confirm that the redundant link is active and available
- Check: Ensure that the Basavanagudi POP is not experiencing any other network issues

Potential Root Causes

Cause	Probability
Microbursts and interface tail drops due to PFC protocol misconfiguration	High
QoS queue starvation due to insufficient buffer space	Medium
Over-subscription of the egress queue	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface egress queue, Expected Output: Queue statistics, including queue depth and drop count, Description: This command will provide information on the egress queue statistics, including queue depth and drop count. This can help identify if the queue is congested and if packets are being dropped.

Command: show qos configuration, Expected Output: QoS configuration, including PFC settings and egress queue configuration, Description: This command will provide information on the QoS configuration, including PFC settings and egress queue configuration. This can help identify if the PFC protocol is misconfigured or if the egress queue configuration is not optimal.
```

```
Command: show buffer statistics, Expected Output: Buffer statistics, including available buffer space, Description: This command will provide information on the available buffer space in the egress queues. This can help identify if the QoS queue starvation is caused by insufficient buffer space.
```

Resolution Steps

Step 1: Step: Verify PFC protocol configuration, Command: show qos configuration | include PFC, Description: Verify that the PFC protocol is correctly configured on the egress queue. If misconfigured, adjust the PFC settings to optimize traffic flow.

Step 2: Step: Adjust egress queue configuration, Command: config qos egress queue buffer-size , Description: Adjust the egress queue configuration to increase the buffer size and optimize traffic flow.

Step 3: Step: Verify buffer space availability, Command: show buffer statistics, Description: Verify that the available buffer space in the egress queues has increased after adjusting the egress queue configuration.

Post-Check Verification

```
Command: show interface egress queue, Expected Output: Queue statistics, including queue depth and drop count, Description: Verify that the egress queue statistics have improved and that packets are no longer being dropped.

Command: show qos configuration, Expected Output: QoS configuration, including PFC settings and egress queue configuration, Description: Verify that the QoS configuration has been updated correctly and that the PFC protocol is correctly configured.

Command: show buffer statistics, Expected Output: Buffer statistics, including available buffer space, Description: Verify that the available buffer space in the egress queues has increased.
```

Roll-back Procedures

- Step: Restore original PFC protocol configuration, Command: config qos egress queue pfc , Description: Restore the original PFC protocol configuration to prevent further packet drops.

- Step: Restore original egress queue configuration, Command: config qos egress queue buffer-size , Description: Restore the original egress queue configuration to prevent further packet drops.